

Ananya Gupta

MS CS @ EPFL · ML Systems & Low-level Optimization · ex-Google SDE III · CUDA / C++ / PyTorch

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EDUCATION

- **EPFL, Lausanne, Switzerland** GPA: 5.56/6.0
MS in Computer Science *Sept. 2024 - Expected 2026*
Coursework: Modern Natural Language Processing, Parallel and High-Performance Computing, Systems for Data Management and Data Science, Mathematics of Data, Machine Learning, Reinforcement Learning, Distributed Algorithms, Concurrent Computing
- **Indian Institute of Technology, BHU, Varanasi, India** GPA: 9.58/10.0
Bachelor of Technology in Electrical Engineering *Jul. 2016 - May 2020*
Director's Gold Medal (top all-round graduate, batch of 1000+)

PUBLICATIONS

- **Tile-Based Decompression in Compiled GPU Engines.** A. Boffa, **A. Gupta**, H. Nicholson, A. Ailamaki. *Accepted, VLDB 2027*. [\[paper\]](#)

RESEARCH PROJECTS

- **Efficient Mixture-of-Experts Training via Asynchronous Methods** EPFL
Master's Thesis, MLO (Prof. Martin Jaggi) & LIONS (Prof. Volkan Cevher) *Sept. 2026 - Ongoing*
 - Investigating **asynchronous and communication-efficient methods** for large-scale **MoE** training, targeting expert load imbalance and GPU underutilization in expert parallelism.
 - Aligned with **Apertus**, the Swiss AI Initiative's open LLM, to inform more efficient training of open models.
- **Bandwidth-Efficient GPU Kernel Fusion via JIT-Compiled CUDA** [\[report\]](#) EPFL
Semester Project, DIAS Lab, Prof. Anastasia Ailamaki | CUDA, C++, Nsight *Feb. 2026 - Ongoing*
 - Co-designed a **modular decompression operator** in Proteus that fuses tile decompression with downstream computation in a single JIT-compiled CUDA kernel, eliminating HBM round-trips and cutting GPU memory-bandwidth use by up to **6×** while matching or beating compressed baselines in runtime.
 - Built a **standalone CUDA micro-benchmark** as a hardware oracle to isolate JIT overhead vs. hand-tuned kernels, characterizing shared-memory pressure, register spilling, and occupancy.
 - Achieved up to **2.9×** **lower GPU memory footprint** at competitive runtime, opening a new region of the memory-runtime Pareto frontier across four query workloads.
- **Extending LUNA (Generative Tissue Reassembly) to Cell Morphologies** [\[report\]](#) EPFL
Semester Project, MLBio Lab, Prof. Maria Brbić | Python, PyTorch, POT *Mar. 2025 - Jul. 2025*
 - Extended **LUNA**, a diffusion model for spatial tissue reconstruction, to a 360K-cell Alzheimer's dataset, reaching **58.9%** Spearman correlation (vs. 44.8% prior best), and added cell morphology via DINOv2/MAE encoders fused through a transformer.
 - Built a shape-guidance step using **Procrustes analysis** and **optimal transport** to align predicted cells with tissue geometry, addressing LUNA's failure on unseen tissue shapes.

WORK EXPERIENCE

- **AI Engineer Intern** Lausanne
AI Team, Nexthink | Python, LangSmith, GraphRAG, Neo4j, FAISS *Aug. 2024 - Jan. 2025*
 - Improved enterprise RAG retrieval across knowledge bases, tickets, and docs; built two LLM-labeled QA datasets and a LangSmith framework benchmarking TF-IDF, BM25, semantic, hybrid, and graph retrieval.

- Designed a **schema-first Knowledge Graph pipeline** (~5K nodes, 12K edges) with parallel extraction and deduplication for multi-hop reasoning.
- Developed a **4-stage RSSR pipeline** with hybrid retrieval and **PageRank-based fused ranking**, beating semantic-only baselines; dynamic query routing improved latency–quality tradeoffs.
- **Software Engineer III** Bangalore, India
Google Search, Google | C++, Java Aug. 2020 - Aug. 2024
 - Co-designed an **LLM-driven agentic workflow** automating code generation, building, and testing for large-scale Search migrations, with evaluation and feedback loops to improve model performance.
 - Contributed to an internal C++ **templating compiler** that compiles one source into **Java (backend) and JavaScript (frontend)** code.
 - Migrated the first Search features to a new microservices architecture (Java, C++), cutting **build time 50%, edit-refresh 75%, and page-reload 68%**.
 - Automated a migration-tracking dashboard, cutting manual prerequisite analysis from **days to minutes**.
 - **Google Research (AI for Social Good)**: Mentored a team of students for an AI assistant for child deliveries in low-resource areas, building **Kaldi ASR models** for partograph documentation and anomaly detection; piloted in a rural maternity centre in Patna, India.

COURSE PROJECTS

- **GPU Kernel Optimization & MPI Scaling for Lattice Boltzmann** EPFL
Parallel and High-Performance Computing | CUDA, MPI, C++ Spring 2026
 - Implemented a D2Q9 BGK LBM solver in **MPI** and **CUDA**: **178.9×** strong-scaling speedup at P=256 (MPI) and **4,755 MLUPS** on a V100 (CUDA).
 - Designed a **fused CUDA kernel** merging collision and streaming, halving memory traffic (288→144 B/cell), doubling occupancy (25%→50%), and reaching 76% of V100 HBM2 bandwidth, guided by roofline analysis and register/block-size profiling.
- **AudioLLM-D: Fine-tuning Audio LLM for Dementia Speech** [code] [report] EPFL
Modern NLP | Python, PyTorch, Hugging Face, PEFT/LoRA Spring 2026
 - Fine-tuned **Qwen2-Audio-7B-Instruct** on DementiaBank with **LoRA** (encoder, projector, LM), adding 8 disfluency tokens and frequency-tiered focal loss for class imbalance.
 - Achieved WER **0.334** and disfluency F1 **0.572**, beating Whisper-large-v3 (0.514) and CrisperWhisper (0.467); ablations identified encoder adaptation as the key bottleneck.

HONOURS AND AWARDS

- **IUSSTF-Viterbi Program** (1 of 15 students from India): summer research on **biomimetic neuromorphic circuits** and spiking neural networks with [Prof. Alice Parker](#), USC (2019).
- Secured an **All India rank of 2318** (0.2 million candidates) in the Joint Entrance Examination (Advanced) for admissions to the Indian Institutes of Technology.

EXTRACURRICULAR

- **Teaching Assistant, EPFL**: [Deep Learning for Autonomous Vehicles](#), [Machine Learning \(CS-433\)](#), and [Introduction to Machine Learning](#).
- **Mentor, Topmate**: 1:1 sessions on applying for Higher Education, interview preparation, and moving towards research.
- **Joint secretary, Robotics Club** (2018–19): Organized robotics events and workshops, and led the college robotics team in competitions.
- **Joint secretary, Social Service Club** (2016–2020): taught underprivileged children; organized medical camps.